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Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 20	
TITLE : JDBC INSERT, UPDATE, and DELETE Operations	

Problem Statement

Develop a Java application to establish connectivity with a database and perform INSERT, UPDATE, and DELETE operations.

Learning Objectives

To perform database modification operations using JDBC.

Exercise

1. **Create a Java application to manage employee records using CRUD operations**

Code:

```
import java.sql.*;
import java.util.Scanner;

public class EmployeeCRUD {

    static String url = "jdbc:mysql://localhost:3306/employee";
    static String user = "root";
    static String password = "mini@12";

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(url, user, password);

            System.out.print("Enter Employee ID: ");
            int id = sc.nextInt();
            sc.nextLine();
```

```
System.out.print("Enter Employee Name: ");
String name = sc.nextLine();

System.out.print("Enter Department: ");
String department = sc.nextLine();

System.out.print("Enter Salary: ");
double salary = sc.nextDouble();

String insert = "INSERT INTO employee VALUES (?, ?, ?, ?)";
PreparedStatement ps = con.prepareStatement(insert);

ps.setInt(1, id);
ps.setString(2, name);
ps.setString(3, department);
ps.setDouble(4, salary);

ps.executeUpdate();
System.out.println("Employee record inserted.");

String select = "SELECT * FROM employee";
Statement stmt = con.createStatement();
ResultSet rs = stmt.executeQuery(select);

System.out.println("\nEmployee Records:");
System.out.println("-----------------------------");

while (rs.next()) {
   System.out.println(
      rs.getInt("employee_id") + " | " +
      rs.getString("name") + " | " +
      rs.getString("department") + " | " +
      rs.getDouble("salary")
   );
}

System.out.print("\nEnter Employee ID to update salary: ");
int updateId = sc.nextInt();

System.out.print("Enter new salary: ");
double newSalary = sc.nextDouble();
```

```
            String update = "UPDATE employee SET salary = ? WHERE
employee_id = ?";
            PreparedStatement psUpdate = con.prepareStatement(update);

            psUpdate.setDouble(1, newSalary);
            psUpdate.setInt(2, updateId);

            psUpdate.executeUpdate();
            System.out.println("Employee record updated.");

            System.out.print("\nEnter Employee ID to delete: ");
            int deleteId = sc.nextInt();

           String delete = "DELETE FROM employee WHERE employee_id = ?";
            PreparedStatement psDelete = con.prepareStatement(delete);

            psDelete.setInt(1, deleteId);

            psDelete.executeUpdate();
            System.out.println("Employee record deleted.");

            con.close();
            sc.close();

        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Output:

```
d=y,address=localhost:62425' '@C:\Users\minis\A
ppData\Local\Temp\cp_7p6f0nbkg00gfn7benzo3fpbn.
argfile' 'EmployeeCRUD'
Enter Employee ID: 292
Enter Employee Name: Mini
Enter Department: CSE
Enter Salary: 9000000
Employee record inserted.

Employee Records:
----------------------------
292 | Mini | CSE | 9000000.0

Enter Employee ID to update salary: 292
Enter new salary: 100000000
Employee record updated.

Enter Employee ID to delete: 292
Employee record deleted.
PS C:\Users\minis\Downloads\javaprograms>
```

2. **Create a Java application to manage student records using CRUD operations. Perform Create, Read, Update, and Delete operations on student details such as roll number, name, course, and marks using a database.**

Code:

```
import java.sql.*;
import java.util.Scanner;

public class StudentCRUD {

    static String url = "jdbc:mysql://localhost:3306/college";
    static String user = "root";
    static String password = "mini@12";

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(url, user, password);
```

```
System.out.print("Enter Roll Number: ");
int rollNo = sc.nextInt();
sc.nextLine();

System.out.print("Enter Name: ");
String name = sc.nextLine();

System.out.print("Enter Course: ");
String course = sc.nextLine();

System.out.print("Enter Marks: ");
double marks = sc.nextDouble();

String insert = "INSERT INTO student VALUES (?, ?, ?, ?)";
PreparedStatement ps = con.prepareStatement(insert);

ps.setInt(1, rollNo);
ps.setString(2, name);
ps.setString(3, course);
ps.setDouble(4, marks);

ps.executeUpdate();
System.out.println("Student record inserted.");

String select = "SELECT * FROM student";
Statement stmt = con.createStatement();
ResultSet rs = stmt.executeQuery(select);

System.out.println("\nStudent Records:");
System.out.println("----------------------------");

while (rs.next()) {
   System.out.println(
      rs.getInt("roll_no") + " | " +
      rs.getString("name") + " | " +
      rs.getString("course") + " | " +
      rs.getDouble("marks")
   );
}

System.out.print("\nEnter Roll Number to update marks: ");
int updateRollNo = sc.nextInt();

System.out.print("Enter new marks: ");
```

```
            double newMarks = sc.nextDouble();

             String update = "UPDATE student SET marks = ? WHERE roll_no =
?";
            PreparedStatement psUpdate = con.prepareStatement(update);

            psUpdate.setDouble(1, newMarks);
            psUpdate.setInt(2, updateRollNo);

            psUpdate.executeUpdate();
            System.out.println("Student record updated.");

            System.out.print("\nEnter Roll Number to delete: ");
            int deleteRollNo = sc.nextInt();

            String delete = "DELETE FROM student WHERE roll_no = ?";
            PreparedStatement psDelete = con.prepareStatement(delete);

            psDelete.setInt(1, deleteRollNo);

            psDelete.executeUpdate();
            System.out.println("Student record deleted.");

            con.close();
            sc.close();

        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Output:

```
d=y,address=localhost:52854' '@C:\Users\minis\A
ppData\Local\Temp\cp_7p6f0nbkg00gfn7benzo3fpbn.
argfile' 'StudentCRUD'
Enter Roll Number: 292
Enter Name: Mini
Enter Course: CSE
Enter Marks: 95
Student record inserted.

Student Records:
----------------------------
292 | Mini | CSE | 95.0

Enter Roll Number to update marks: 292
Enter new marks: 98
Student record updated.

Enter Roll Number to delete: 292
Student record deleted.
PS C:\Users\minis\Downloads\javaprograms>
```

Conclusion

In this assignment, we used JDBC to connect Java applications with a database and performed Create, Read, Update, and Delete (CRUD) operations on student and employee records. We learned how to insert, retrieve, update, and delete data using SQL queries. The successful execution of these operations helped us understand database connectivity and data management using JDBC.